MM : 720

TEST - I

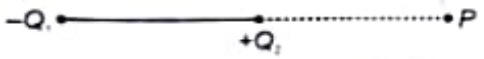
Time : 3 Hrs.

[PHYSICS]

Choose the correct answer :

1. Two point charges +0.03 μC and −0.02 μC are located [redacted] cm apart. The distance of the point on the line joining the two charges where electric potential is zero is (Take the potential at infinity to be zero)
   (1) 9 cm from positive charge
   (2) 9 cm from negative charge
   (3) 45 cm from positive charge
   (4) Both (1) and (3)

2. Two point charges, $-Q_1$ and $+Q_2$, are placed along a straight line as shown in the diagram. If $\vec{E}_1$ and $\vec{E}_2$ are the electric field vectors at point P due to charges $-Q_1$ and $+Q_2$, respectively, then correct option is

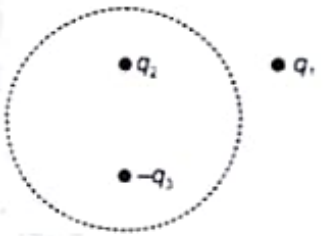


   (1) $\vec{E}_1 \times \vec{E}_2 = 0$ (2) $\vec{E}_1 \cdot \vec{E}_2 < 0$
   (3) Both (1) and (2) (4) $|\vec{E}_1| = |\vec{E}_2|$

3. A cylinder of length L and radius r has its axis coincident with x-axis. The electric field in this region is $\vec{E} = (100\hat{i} + 100\hat{j})$. The magnitude of flux entering through flat surface of the cylinder is
   (1) $100\sqrt{2}\pi r^2$
   (2) Zero
   (3) $200\pi r^2$
   (4) $100\pi r^2$

4. Select the correct statement regarding Coulomb's law.
   (1) Coulomb's law agrees with the Newton's third law
   (2) Coulomb's law is a quantitative statement about the force between two point charges
   (3) Coulomb's law follows inverse square law relation with distance
   (4) All of the above

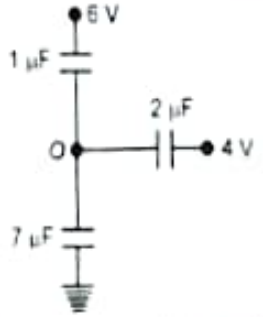
5. Consider the charge configuration and spherical gaussian surface as shown in the figure.



   Statement-A : Electric flux of the electric field over the surface is due to q_2 and $-q_3$ only.
   Statement-B : Electric field at the gaussian surface will be due to q_1 only.
   Select the correct option.
   (1) Only statement A is [redacted]
   (2) Only statement B is [redacted]
   (3) Both statements A and B are [redacted]
   (4) Both statements A and B are incorrect

Space for Rough Work

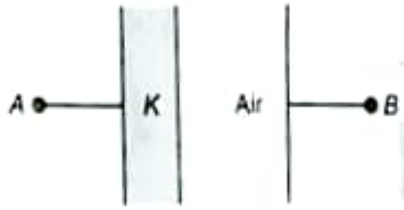
6. Three capacitors are connected as shown in the figure. The potential at point O is equal to



(1) [redacted] V (2) [redacted] V

(3) [redacted] V (4) 2.0 V

7. A parallel plate air capacitor has capacitance C. If one-third of space between the plates is filled with dielectric of dielectric constant K as shown in the figure then the new capacitance C' will be



(1) $C' = \frac{3C}{1+K}$ (2) $C' = \frac{C}{2}(1+K)$

(3) $C' = \frac{3}{2}\left[K + \frac{1}{2}\right]$ (4) $C' = C\left(1 + \frac{K}{2}\right)$

8. Two charges $+q$ and $-q$, each of mass m are revolving in a circle of radius R under mutual electrostatic force. Total energy of the system is

(1) $\frac{Kq^2}{2R}$ (2) $\frac{-Kq^2}{4R}$

(3) $\frac{-Kq^2}{R}$ (4) $\frac{Kq^2}{8R}$

9. If $\oint \vec{E} \cdot \vec{ds} = 0$ over a surface, then

(1) All the charges must necessarily be inside the surface

(2) The number of flux lines entering the surface must be equal to the number of flux lines leaving it

(3) The net charge enclosed within that surface is zero

(4) Both (2) and (3)

10. Consider an isolated conducting sphere having radius same as the Earth (6400 km). The capacitance of this sphere is approximately equal to

(1) Zero

(2) Infinite

(3) 711 μF

(4) 6.4×10^6 F

11. A charge Q is placed at the corner of a cube. Flux through one of the faces of the cube not touching the charge is

(1) $\frac{Q}{\varepsilon_0}$ (2) $\frac{Q}{8\varepsilon_0}$

(3) $\frac{Q}{24\varepsilon_0}$ (4) Zero

12. A charge $q_1 =$ [redacted] μC is located at the origin and a charge $q_2 =$ [redacted] μC is located at (0, 3) m. Change in potential energy of the system if a charge $q_3 =$ [redacted] μC moves from infinity to point P, whose coordinates are (4, 0) m, is

(1) 13.5×10^{-3} J (2) 4.50×10^{-2} J

(3) 40.5×10^{-3} J (4) 22.5×10^{-4} J

Space for Rough Work

13. If a charge q is placed at each vertex except one vertex of a regular polygon of n sides, then the magnitude of net electric field at its centre, (assume the distance of centre from each vertex to be r) is

(1) $\frac{q}{n4\pi\varepsilon_0 r^2}$ (2) $\frac{nq}{4\pi\varepsilon_0 r^2}$

(3) $\frac{(n+1)q}{4\pi r^2}$ (4) $\frac{q}{4\pi\varepsilon_0 r^2}$

14. An electric dipole consists of two opposite charges each of magnitude 5 μC separated by a distance 1 cm. The dipole is placed in an external field of 10^5 N/C. The maximum torque on the dipole is

(1) 25×10^{-2} Nm (2) 5×10^{-3} Nm

(3) 0.5×10^{-1} Nm (4) 5×10^{-6} Nm

15. A long wire with a linear charge density λ per unit length passes through an imaginary cuboid of edges $\left(a, a, \frac{a}{2}\right)$. Maximum possible flux of the electric field through the cuboid is

(1) $\frac{\sqrt{3}a\lambda}{\varepsilon_0}$ (2) $\frac{a\lambda}{\varepsilon_0}$

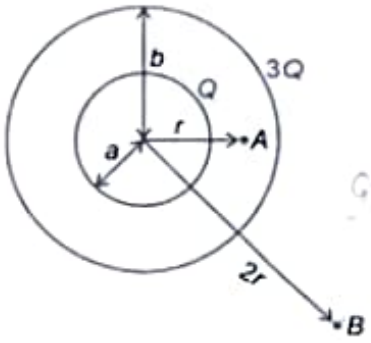
(3) $\frac{3}{2}\frac{a\lambda}{\varepsilon_0}$ (4) $\frac{3a\lambda}{\varepsilon_0}$

16. A charge of 5 C is moved in an electric field of a fixed charge distribution from point A to another point B slowly. The work done by external agent in doing so is [redacted] J, potential difference $V_A - V_B$ is

(1) 20 V (2) –20 V

(3) 500 V (4) –500 V

17. The electric field intensity at points A and B, in the arrangement of two concentric shells, are in the ratio



(1) $\frac{1}{2}$ (2) 4

(3) $\frac{a}{b}$ (4) 1

18. A sphere of radius [redacted] m is placed concentrically within a spherical shell of radius [redacted] m. The inner sphere has potential of 2 V and the other sphere is earthed. The charge (in coulombs) on the [redacted] sphere is $\left(\text{Where } K = \frac{1}{4\pi\varepsilon_0}\right)$.

(1) $\frac{32}{K}$ (2) $\frac{12}{K}$

(3) $\frac{24}{K}$ (4) $\frac{6}{K}$

19. The energy stored and capacitance of a charged parallel plate capacitor are U and C respectively. Now a dielectric slab of $\varepsilon_r = 4$ is completely inserted in it, then energy stored and capacitance becomes (Assuming charges on plates remain constant)

(1) $4U, 4C$ (2) $\frac{U}{4}, 4C$

(3) U, C (4) $U, 4C$

Space for Rough Work

20. The magnitude of electric field due to a small dipole of dipole moment P at any point that lies on the line making an angle θ with the axis of dipole at distance r is given by

(1) $E = \frac{1}{4\pi\varepsilon_0}\frac{P}{r^3}\sqrt{\cos^2\theta - 1}$

(2) $E = \frac{1}{4\pi\varepsilon_0}\frac{2P}{r^3}\sqrt{\cos^2\theta - 1}$

(3) $E = \frac{1}{4\pi\varepsilon_0}\frac{P}{r^3}\sqrt{3\sin^2\theta + 1}$

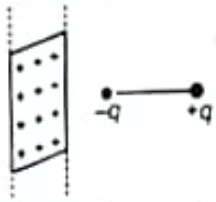
(4) $E = \frac{1}{4\pi\varepsilon_0}\frac{P}{r^3}\sqrt{3\cos^2\theta + 1}$

21. A very long uniformly charged thin rod is placed along y-axis with its one end at origin and it extends along + y-axis. The charge per unit length of the rod is λ. then electric field at $y = -a$ on y-axis is

(1) $\frac{\lambda\hat{j}}{4\pi\varepsilon_0 a}$ (2) $\frac{-\lambda\hat{j}}{4\pi\varepsilon_0 a^2}$

(3) $\frac{-\lambda\hat{j}}{4\pi\varepsilon_0 a}$ (4) $\frac{\lambda\hat{j}}{4\pi\varepsilon_0 a}$

22. An electric dipole is placed at a certain distance from an infinite plane sheet having uniform positive charge density σ. Based on above information choose the correct option from the following.



(1) Torque on the dipole is zero and net force is directed away from the sheet

(2) Torque on the dipole is zero and net force acts towards the sheet

(3) Potential energy of dipole is minimum and torque is zero

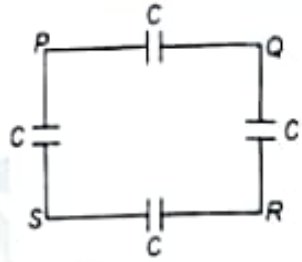
(4) Potential energy of dipole is maximum, and torque is also maximum

23. A parallel plate capacitor of capacitance 1 μF is charged to a potential difference of [redacted] V. The distance between the plates is [redacted] mm. The energy density between plates of capacitor is

(1) 100 J/m^3 (2) 4.42 J/m^3

(3) 52.75 J/m^3 (4) 16.85 J/m^3

24. Four capacitors, each of capacitance C are connected as shown in figure. The ratio of equivalent capacitance between points [redacted] to the capacitance between points [redacted] will be



(1) $\frac{1}{2}$ (2) $\frac{4}{3}$

(3) $\frac{3}{2}$ (4) $\frac{3}{4}$

25. Two metallic spheres of radii r and R ($r < R$) are placed far away from each other and are connected by a thin conducting wire. Now a charge Q is given to the larger sphere then final charge on smaller sphere at electrostatic equilibrium, will be

(1) Q (2) $\frac{Qr}{R}$

(3) $\frac{Qr}{R+r}$ (4) $\frac{QR}{r+R}$

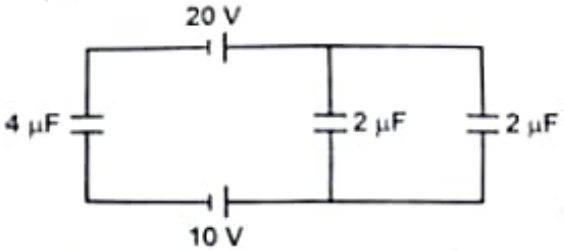
Space for Rough Work

26. A point charge q is placed inside a conducting spherical shell of inner radius [redacted] and outer radius [redacted] at a distance R from the centre of the shell. The electric potential at the centre of shell will be (Here $k = \frac{1}{4\pi\varepsilon_0}$)

(1) $\frac{kq}{2R}$ (2) $\frac{4kq}{3R}$

(3) $\frac{5kq}{6R}$ (4) $\frac{2kq}{3R}$

27. In the circuit shown in figure, charge stored in [redacted] μF capacitor is



(1) 20 μC (2) 40 μC

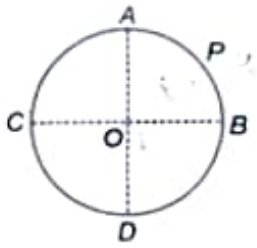
(3) 60 μC (4) 10 μC

28. Two small identically charged spheres are suspended at common point by strings of equal length. The strings make an angle of 30° with each other. When suspended in a liquid of density 0.6 g/cc, the angle remains same. If density of sphere = 1.8 g/cc, then dielectric constant of liquid is

(1) $K = 2$ (2) $K = 3$

(3) $K = \frac{3}{2}$ (4) $K = \frac{4}{3}$

29. A thin conducting ring of radius R is given a charge $+Q$. The electric field at the centre O of the ring due to the charge on the part APB of the ring is E. The electric field at the centre due to the charge at the part ACD of the ring will be



(1) Zero (2) $2E$ along OP

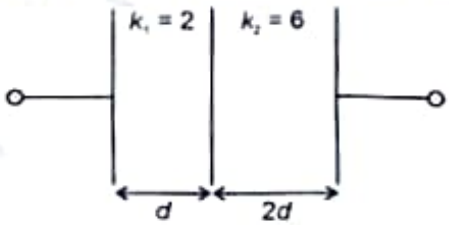
(3) E along OP (4) $\sqrt{2}E$ along OB

30. The capacitance of system of two concentric spherical shells having radius a and b ($a < b$) is C_1, when outer sphere is earthed while the capacitance of system is C_2 when inner sphere is earthed. Then [redacted] will be

(1) 1 (2) $\frac{a^2}{b^2}$

(3) $\frac{a}{b}$ (4) $\frac{a+b}{a-b}$

31. A parallel plate capacitor has two layers of dielectrics as shown in figure. This capacitor is connected across a battery, then the ratio of potential difference across the dielectric layers is



(1) $\frac{3}{2}$ (2) $\frac{1}{2}$

(3) $\frac{4}{3}$ (4) $\frac{1}{3}$

Space for Rough Work

32. A capacitor is connected to a battery. The force of attraction between the plates, when the separation between them is [redacted].

(1) Remains the same (2) Becomes 8 times

(3) Becomes 4 times (4) Becomes 2 times

33. If [redacted] electrons are [redacted] from a neutral metallic sphere, then charge on the sphere will be

(1) +16 mC (2) −1.6 mC

(3) +1.6 mC (4) +3.2 mC

34. Given below are two statements :

Statement-I : The number of electric field lines originating or terminating on a charge is inversely proportional to the [redacted].

Statement-II : The electrostatic field lines [redacted] form any closed loop due to conservative nature of electric field.

In the light of the above statements, choose the **most appropriate** answer from the options given below.

(1) Statement I is correct but Statement II is incorrect

(2) Statement I is incorrect but Statement II is correct

(3) Both Statement I and Statement II are correct

(4) Both Statement I and Statement II are incorrect

35. A short electric dipole has a dipole moment of 6×10^{-9} C m. The electric potential due to the dipole at a point at a distance of 0.5 m from the centre of the dipole, situated on a line making an angle of [redacted] with the dipole is

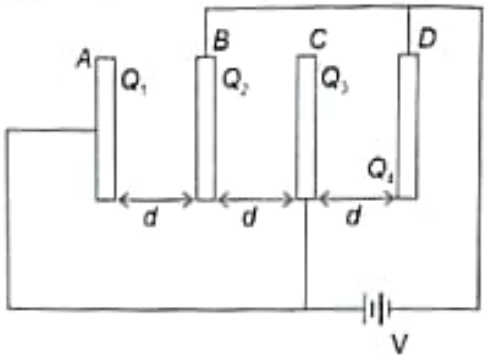
(1) 54 V

(2) 72 V

(3) 108 V

(4) Zero

36. For the given arrangement of four large parallel conducting plates, net charges on plates A, B, C and D are Q_1, Q_2, Q_3 and Q_4 respectively given in column-I and their values given in column-II. Match column-I and column-II (Area of each plate is A)



Select the most appropriate option.

	Column-I		Column-II
(A)	Q_1	(P)	$\frac{-A\varepsilon_0 V}{d}$
(B)	Q_2	(Q)	$\frac{2A\varepsilon_0 V}{d}$
(C)	Q_3	(R)	$\frac{A\varepsilon_0 V}{d}$
(D)	Q_4	(S)	$\frac{-2A\varepsilon_0 V}{d}$

(1) A(P), B(Q), C(R), D(S)

(2) A(R), B(S), C(P), D(Q)

(3) A(R), B(S), C(Q), D(P)

(4) A(R), B(Q), C(P), D(S)

Space for Rough Work

37. A non-conducting spherical shell of radius r having a charge q_1 is placed at a center to center separation distance of $3r$ from another charged thin non-conducting spherical shell of radius r having charge q_2. If charge distributions are considered uniform, then the total electrostatic potential energy of the system is

(1) $\frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{r}+\frac{q_2}{r}+\frac{q_1q_2}{3r}\right]$

(2) $\frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{3r}$

(3) $\frac{1}{4\pi\varepsilon_0}\left[\frac{q_1^2}{2r}+\frac{q_2^2}{2r}+\frac{q_1q_2}{3r}\right]$

(4) $\frac{1}{4\pi\varepsilon_0}\left[\frac{q_1^2}{r}+\frac{q_2^2}{r}+\frac{q_1q_2}{3r}\right]$

38. Consider the following statements.

Statement A : Coulomb's force is central force as well as [redacted] in nature.

Statement B : Coulomb's force may be [redacted]

Statement C: Coulomb's force follows superposition principle and [redacted]

Choose the correct option from the given below.

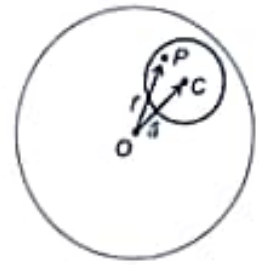
(1) Only statements A and B are correct

(2) Only statements B and C are correct

(3) Only statements A and C are correct

(4) All the statements A, B and C are correct

39. Consider any point P inside a cavity in a uniformly charged sphere having charge density ρ as shown. The position vector of P is $\vec{r}$ and of centre of cavity (C) is $\vec{a}$. Then electric field at point $P(\vec{E}_p)$ is given by



(1) $\vec{E}_p = 0$

(2) $\vec{E}_p = \frac{\rho\vec{a}}{3\varepsilon_0}$

(3) $\vec{E}_p = \frac{\rho\vec{r}}{3\varepsilon_0}$

(4) $\vec{E}_p = \frac{\rho(\vec{a}+\vec{r})}{4\varepsilon_0}$

40. A 3 μF capacitor is charged to a potential of [redacted] V and 2 μF capacitor is charged to [redacted] V. The capacitors are then connected in parallel with plates of opposite polarity joined together. Amount of charge that will flow when the plates are so connected,

(1) 200 μC (2) 600 μC

(3) Zero (4) 100 μC

41. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Work done in moving a positive charge on an equipotential surface is [redacted].

Reason (R): Electric potential difference between two points is defined as the work done by an external force to move a [redacted] from one point to another within an electric field.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of Assertion (A).

(3) Assertion (A) is true, but Reason (R) is false.

(4) Assertion (A) is false, but Reason (R) is true.

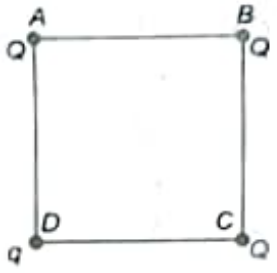
Space for Rough Work

42. If the plates of a parallel plate capacitor connected to a battery are moved close to each other, then
(A) The charge stored in it decreases
(B) The energy stored in it decreases
(C) Its capacitance increases
(D) Product of charge and voltage increases
The correct statements are
(1) A and B only (2) A, B and C only
(3) B, C and D only (4) C and D only

43. If a point charge is taken from [redacted] potential to [redacted] potential, then
(1) Potential energy remains constant
(2) Potential energy always increases
(3) Potential energy always decreases
(4) Potential energy may increase or decrease

44. Two equal point charges, placed [redacted] mm apart, are released from rest. Initial acceleration of one of them is [redacted] m/s² and its mass is 5×10^{-5} g. If initial acceleration of other charge is [redacted] m/s², then mass and magnitude of other charge are
(1) 4×10^{-5} kg, 33 μC (2) 4×10^{-5} g, 33.3 pC
(3) 4×10^{-5} kg, 33.3 μC (4) 4×10^{-5} g, 11.1 pC

45. Three charges each of charge Q are placed at vertices of a square $ABCD$, as shown in figure. The value of charge q placed at remaining vertex (in terms of Q), such that force on charge Q placed at B is zero, is



(1) $q = -\sqrt{2}Q$ (2) $q = -2\sqrt{2}Q$
(3) $q = +2Q$ (4) $q = \left(-2 + \sqrt{2}\right)Q$

[CHEMISTRY]

46. O_2 gas will be evolved as a product of electrolysis of
(a) An aqueous solution of $AgNO_3$ using sliver electrodes.
(b) An aqueous solution of $AgNO_3$ using platinum electrodes.
(c) A dilute solution of H_2SO_4 using platinum electrodes.
(d) A high concentration solution of H_2SO_4 using platinum electrodes.
Choose the correct answer from the options given below.
(1) (b) and (d) only (2) (a) and (c) only
(3) (b) and (c) only (4) (a) and (d) only

47. If resistance of conductivity cell filled with NaCl solution is 50 Ω and conductivity of solution is 0.015 S cm^{-1} then the cell constant of conductivity cell will be
(1) 1.25 cm^{-1}
(2) 0.5 cm^{-1}
(3) 0.75 cm^{-1}
(4) 1.50 cm^{-1}

48. The mass of a non-volatile, non-electrolyte (molar mass = 50 g mol^{-1}) needed to be dissolved in 114 g octane to reduce its vapour pressure to 70%, is
(1) 37.5 g (2) 70 g
(3) 150 g (4) 21.42 g

Space for Rough Work

49. Ethylene glycol ($C_2H_6O_2$) is used as an antifreeze in cold climate. Mass of ethylene glycol which should be added to 2 kg of water to prevent it from freezing at –4°C will be [K_f for water = 1.86 K kg mol^{-1})
(1) 266.6 g (2) 304.6 g
(3) 400.0 g (4) 204.3 g

50. Which of the following statements is incorrect regarding a solution of two components A and B exhibiting negative deviation from ideal behaviour?
(1) Intermolecular attractive forces between A-A and B-B are weaker than those between A-B.
(2) $\Delta_{mix}H < 0$ at constant T and P
(3) $\Delta_{mix}V < 0$ at constant T and P
(4) $\Delta S < 0$

51. Given below are two statements.
Statement-I : Blood cells retain their normal shape in solution which are isotonic to blood.
Statement-II : The values of K_f and K_b is directly proportional to molar mass of solute.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Statement I is incorrect but Statement II is correct
(2) Statement I is correct but Statement II is incorrect
(3) Both Statement I and Statement II are incorrect
(4) Both Statement I and Statement II are correct

52. Vapour pressure of chloroform ($CHCl_3$) and dichloromethane (CH_2Cl_2) at 25°C are 210 mm Hg and 415 mm Hg respectively. Vapour pressure of the solution obtained by mixing 26 g of $CHCl_3$ and 40 g of CH_2Cl_2 at the same temperature will be nearly
(1) 186.22 mm of Hg (2) 260.30 mm of Hg
(3) 236.81 mm of Hg (4) 350.24 mm of Hg

53. The following solutions were prepared at 25°C by dissolving 20 g of glucose ($C_6H_{12}O_6$) in 500 mL of water (P_1), 20 g of urea (CH_4N_2O) in 500 mL of water (P_2) and 20 g of sucrose ($C_{12}H_{22}O_{11}$) in 500 mL of water (P_3). The correct option for the increasing order of osmotic pressure of these solutions is
(1) $P_2 < P_1 < P_3$ (2) $P_3 < P_1 < P_2$
(3) $P_3 < P_2 < P_1$ (4) $P_1 < P_3 < P_2$

54. A 0.1 molal aqueous solution of a weak monoprotic acid is 40% ionised. If K_f for water is 1.86 K kg/mol, the freezing point of the solution will be
(1) –0.13°C (2) –0.54°C
(3) –0.26°C (4) –0.36°C

55. The pressure of H_2 required to make the potential of H_2 electrode zero in pure water at 298 K is
(1) 10^{-12} atm (2) 10^{-16} atm
(3) 10^{-7} atm (4) 10^{-14} atm

56. If the E°_{cell} for a given reaction has a positive value, which of the following gives the correct relationships for the values of ΔG° and K_{eq} ?
(1) $\Delta G^\circ > 0$; $K_{eq} < 1$ (2) $\Delta G^\circ > 0$; $K_{eq} > 1$
(3) $\Delta G^\circ < 0$; $K_{eq} > 1$ (4) $\Delta G^\circ < 0$; $K_{eq} < 1$

57. Match List-I with List-II.

	List-I (Cell)		List-II (Reaction at Cathode)
a.	Dry cell	(i)	$O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$
b.	Mercury cell	(ii)	$MnO_2 + NH_4^+ + e^- \rightarrow MnO(OH) + NH_3$
c.	Lead storage battery (on discharging)	(iii)	$HgO + H_2O + 2e^- \rightarrow Hg(l) + 2OH^-$
d.	$H_2 - O_2$ fuel cell	(iv)	$PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$

Space for Rough Work

Choose the correct answer from the options given below

(1) a(iii), b(ii), c(iv), d(i) (2) a(ii), b(iii), c(iv), d(i)

(3) a(iv), b(ii), c(i), d(iii) (4) a(i), b(iii), c(ii), d(iv)

58. Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion (A) : For mercury cell, the cell potential remains constant during its life.

Reason (R) : In mercury cell, the overall reaction does not involve any ion in solution whose concentration can change during its life time.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both A and R are true but R is not the correct explanation of A

(2) A is true but R is false

(3) A is false but R is true

(4) Both A and R are true and R is the correct explanation of A

59. Consider the following statements

(a) The electric charge for electrode deposition of one gram equivalent of a substance is nearly 96500 coulombs.

(b) The function of salt bridge is to allow the movement of ions from one solution to other without mixing of the two solutions.

(c) In corrosion, a metal is oxidised by loss of electrons to oxygen.

(d) Aluminium metal is produced by electrolysis of molten aluminium oxide in the presence of cryolite.

(e) Copper sulphate solutions can be stored in zinc pot.

Choose the correct statements

(1) (a), (b) and (c) only

(2) (b), (c) and (d) only

(3) (c), (d) and (e) only

(4) (a), (b), (c) and (d) only

60. Consider the following cell reaction

$Sn(s) + Cu^{2+}(0.01M) \rightarrow Sn^{2+}(0.001M) + Cu(s)$

The Gibbs free energy change (in kJ) for the above reaction at 298 K is approximately

[Given : $E^{\circ}_{Cu^{2+}/Cu} = 0.34$ V; $E^{\circ}_{Sn^{2+}/Sn} = -0.14$ V]

(1) −98.43 (2) −196.86

(3) −44.39 (4) −11.9

61. The correct order of molar conductance at infinite dilution of NaCl, KCl, HCl and NaOH is

(1) NaCl > KCl > NaOH > HCl

(2) HCl > NaOH > KCl > NaCl

(3) NaOH > HCl > KCl > NaCl

(4) KCl > NaCl > HCl > NaOH

62. If conductivity of 10^{-3} M CH_3COOH solution is 1.85×10^{-5} S cm^{-1} and limiting molar conductivity of H^+ and CH_3COO^- ion is 349.6 S cm^2 mol^{-1} and 40.9 S cm^2 mol^{-1} respectively. The degree of dissociation for 10^{-3} M CH_3COOH will be

(1) 9.1% (2) 2.3%

(3) 11.2% (4) 4.7%

63. Match List-I with List-II.

	List-I (Material)		List-II (Conductivity/S m^{-1})
a.	Glass	(i)	6.2×10^3
b.	CuO	(ii)	1×10^{-7}
c.	Teflon	(iii)	1×10^{-16}
d.	Silver	(iv)	1×10^{-18}

Choose the correct answer from the options given below.

(1) a(ii), b(iii), c(i), d(iv) (2) a(i), b(ii), c(iii), d(iv)

(3) a(iv), b(ii), c(iii), d(i) (4) a(iii), b(ii), c(iv), d(i)

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64. The conductivity of electrolytic (ionic) solutions depends on:

(a) The nature of the electrolyte added
(b) Size of the ions produced and their solvation
(c) The nature of the solvent and its viscosity
(d) Concentration of the electrolyte
(e) Temperature

Correct statements are

(1) (a), (b) and (c) only
(2) (c) and (d) only
(3) (a), (b), (d) and (e) only
(4) (a), (b), (c), (d) and (e)

65. Following limiting molar conductivities are given as:

$\Lambda_m^\circ(H_2SO_4) = x\ S\ cm^2\ mol^{-1}$

$\Lambda_m^\circ(Na_2SO_4) = y\ S\ cm^2\ mol^{-1}$

$\Lambda_m^\circ(CH_3COONa) = z\ S\ cm^2\ mol^{-1}$

$\Lambda_m^\circ\left(\text{in } S\ cm^2\ mol^{-1}\right)$ for CH_3COOH will be

(1) $z + \frac{(x-y)}{2}$ (2) $z + x - y$

(3) $z + y - x$ (4) $x - y + 2z$

66. During the electrolysis of molten sodium chloride, the time required to produce 2.24 liters of chlorine gas at STP using a current of 3 amperes is nearly

(1) 214 minutes (2) 107 minutes
(3) 55 minutes (4) 320 minutes

67. In the electrolysis of aqueous NaCl when Pt electrode is taken, H_2 gas is liberated at cathode while with Hg cathode it forms sodium amalgam. The reason for this is

(1) Concentration of H^+ ions is more when Pt electrode is taken
(2) Hg is more inert than Pt
(3) Na is dissolved in Hg while it does not dissolve in Pt
(4) More voltage is required to reduce H^+ at Hg than at Pt

68. Standard electrode potential data are useful for understanding the suitability of an oxidant in a redox titration. Some of half cell reactions and their standard potential are given below.

$Ce^{4+} + e^- \rightarrow Ce^{3+}$ $E^\circ = 1.61\ V$

$Br_2 + 2e^- \rightarrow 2Br^-$ $E^\circ = 1.09\ V$

$Fe^{3+} + e^- \rightarrow Fe^{2+}$ $E^\circ = 0.77\ V$

$Cl_2 + 2e^- \rightarrow 2Cl^-$ $E^\circ = 1.36\ V$

For quantitative estimation of aqueous $FeSO_4$ which of the following statements is incorrect?

(1) Ce^{4+} can be used in H_2SO_4 medium
(2) Ce^{4+} can be used in HCl medium
(3) Ce^{4+} can oxidise Fe^{2+} quantitatively
(4) Ce^{4+} cannot be used in HBr medium

69. The efficiency of a fuel cell is given by

(1) $(\Delta S/\Delta G) \times 100$
(2) $(\Delta H/\Delta G) \times 100$
(3) $(\Delta G/\Delta S) \times 100$
(4) $(\Delta G/\Delta H) \times 100$

70. Consider the following cell reaction

$2Fe(s) + O_2(g) + 4H^+(aq) \rightarrow 2Fe^{2+}(aq) + 2H_2O(l)$

$E^\circ = 1.67\ V$

At $[Fe^{2+}] = 10^{-2}$ M, $p(O_2) = 0.1$ atm and pH = 2, the cell potential at 25°C is

(1) 1.47 V
(2) 1.77 V
(3) 1.59 V
(4) 1.53 V

Space for Rough Work

71. Given below are two statements :

Statement-I : For weak electrolytes, Λ_m cannot be obtained by extrapolation of Λ_m to zero concentration. (In graph plotted between Λ_m and $C^{1/2}$)

Statement-II : For strong electrolytes, Λ_m increases slowly with dilution and can be represented by the equation.

$\Lambda_m = \Lambda_m^\circ - A\,C^{1/2}$

Value of A is same for NaCl and KCl

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Statement I is incorrect but Statement II is correct

(2) Statement I is correct but Statement II is incorrect

(3) Both Statement I and Statement II are incorrect

(4) Both Statement I and Statement II are correct

72. A solution of copper sulphate is electrolysed for 20 minutes with a current of 1.5 amperes. The mass of copper deposited at cathode is (Given : Molar mass of Cu = 63.5 g mol^{-1}, 1 F = 96487 C mol^{-1})

(1) 0.2938 g (2) 0.5923 g

(3) 1.7018 g (4) 2.4036 g

73. Match List-I with List-II.

	List-I (Conversion)		List-II (Number of Faraday required)
a.	2 mol of H_2O to O_2	(i)	2 F
b.	1 mol of $Cr_2O_7^{2-}$ to Cr^{3+}	(ii)	3 F
c.	1.5 mol of Ba from molten $BaCl_2$	(iii)	6 F
d.	1 mol of Fe_2O_3 to FeO	(iv)	4 F

Choose the correct answer from the options given below.

(1) a(iv), b(iii), c(i), d(ii) (2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(iv), c(ii), d(i) (4) a(i), b(iii), c(ii), d(iv)

74. Standard electrode potential for Cu^{2+}/Cu^+ couple is +0.16 V and that for the Cr^{3+}/Cr couple is −0.74 V. These two couples in their standard state are connected to make a cell. The cell potential will be

(1) +0.90 V (2) −0.58 V

(3) +0.58 V (4) −0.90 V

75. The standard electrode potential (E°) values of Cr^{3+}/Cr, Mg^{2+}/Mg, Cu^{2+}/Cu and K^+/K are −0.74 V, − 2.36 V, 0.34 V and −2.93 V, respectively. The correct order of reducing power of the metals is

(1) K > Mg > Cr > Cu (2) Mg > K > Cr > Cu

(3) Cr > Cu > K > Mg (4) Cu > Cr > Mg > K

76. Which of the following aqueous solution will exhibit highest boiling point? (Consider complete dissociation of electrolytes)

(1) 0.01 m K_2SO_4 (2) 0.015 m glucose

(3) 0.01 m $NaNO_3$ (4) 0.01 m urea

77. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): For a binary liquid solutions forming a minimum-boiling azeotrope, the boiling point of the azeotrope is lower than that of both pure component.

Reason (R): For minimum boiling azeotrope, the total vapour pressure of the solution becomes greater than the vapour pressure of each pure component at the same temperature.

In the light of the above statements, choose the correct answer from the options given below.

(1) (A) is true but (R) is false

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

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78. From the colligative properties of solution, which one is the best method for the determination of molecular weight of proteins and polymers?
(1) Relative lowering in vapour pressure
(2) Osmotic pressure
(3) Elevation in boiling point
(4) Lowering in freezing point

79. In the depression of freezing point experiment for NaCl (aq), it is found that the
(a) Vapour pressure of the solution is less than that of pure solvent at constant temperature.
(b) Vapour pressure of the solution is more than that of pure solvent at constant temperature.
(c) Both solvent and solute molecules solidify at the freezing point.
(d) Only solvent molecules solidify at the freezing point.
The correct statement(s) is/are
(1) (a) only (2) (b) and (c) only
(3) (a) and (d) only (4) (b) and (d) only

80. Given below are two statements:
Statement-I : To increase the solubility of CO_2 in soft drinks and soda water, the bottle is sealed under high pressure.
Statement-II : At high altitudes the partial pressure of oxygen is less than that at the ground level.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Statement I is incorrect but Statement II is correct
(2) Statement I is correct but Statement II is incorrect
(3) Both Statement I and Statement II are incorrect
(4) Both Statement I and Statement II are correct

81. We have three aqueous solutions of KCl labelled as A, B and C with concentration 0.1 m, 0.01 m and 0.001 m respectively. The value of van't Hoff factor (i) for these solutions will be in the order
(1) $i_A = i_B = i_C$ (2) $i_A < i_B < i_C$
(3) $i_C < i_B < i_A$ (4) $i_A < i_C < i_B$

82. Match List-I with List-II.

	List-I (Aqueous solution and degree of dissociation)		List-II (Value of van't Hoff factor)
a.	$Al_2(SO_4)_3$, $\alpha = 60°$	(i)	1.9
b.	LiCl, $\alpha = 90°$	(ii)	3.1
c.	Na_3PO_4, $\alpha = 70°$	(iii)	1.5
d.	$BaSO_4$, $\alpha = 50°$	(iv)	3.4

Choose the correct match.
(1) a(iv), b(i), c(ii), d(iii)
(2) a(iii), b(ii), c(iv), d(i)
(3) a(i), b(ii), c(iii), d(iv)
(4) a(ii), b(iii), c(iv), d(i)

83. Benzene and naphthalene form an ideal solution at room temperature. For this process, the true statements/relations are
(a) $\Delta H = 0$ (b) $\Delta S_{surrounding} = 0$
(c) ΔS_{system} is positive (d) ΔG is positive
Choose the correct option.
(1) (a) and (b) only
(2) (c) and (d) only
(3) (a), (b) and (d) only
(4) (a), (b) and (c) only

Space for Rough Work

84. On dissolving a non-volatile non-electrolytic solute in benzene, vapour pressure decreases from 650 mm to 630 mm Hg. The depression of freezing point of benzene (in K) upon addition of the solute is [Given that : Molal freezing point depression constant of benzene is 5.12 K kg mol^{-1}]
(1) 2.08 K (2) 1.03 K
(3) 1.80 K (4) 2.53 K

85. Consider the given solutions
(a) Ethanol + water (b) Water + nitric acid
(c) Acetone + chloroform (d) Ethanol + acetone
Choose the correct option containing solutions that exhibit negative deviation from ideal behaviours.
(1) (a) and (b) only (2) (b) and (c) only
(3) (c) and (d) only (4) (a) and (d) only

86. In a sample of air saturated with water vapour, the mole fraction of water vapour is 0.02. If the total pressure of the saturated air is 1.5 atm, the partial pressure of dry air is
(1) 1.53 atm (2) 1.47 atm
(3) 1.43 atm (4) 1.45 atm

87. The Henry's Law constant (K_H) values of the gases (P, Q, R) in water are 155, 4×10^{-5} and 45 kbar respectively. The solubility of these gases in water follow the order
(1) P > Q > R (2) R > P > Q
(3) Q > R > P (4) Q > P > R

88. How many grams of concentrated nitric acid solution should be used to prepare 300 ml of 2.5 M HNO_3? [The concentrated acid is 70% (w/w) HNO_3]
(1) 67.5 g conc. HNO_3 (2) 54.5 g conc. HNO_3
(3) 45.8 g conc. HNO_3 (4) 70.0 g conc. HNO_3

89. If the mass ratio of methanol and water in a solution is 4 : 9 then the mole fraction of methanol in the solution will be
(1) 0.4 (2) 0.1
(3) 0.6 (4) 0.2

90. Match List-I with List-II.

	List-I (Example)		List-II (Type of Solution)
a.	Smoke	(i)	Solid in solid
b.	Amalgam of mercury with sodium	(ii)	Liquid in gas
c.	Alloys	(iii)	Solid in gas
d.	Humidity	(iv)	Liquid in solid

Choose the correct answer from the options given below.
(1) a(ii), b(iv), c(i), d(iii) (2) a(iv), b(iii), c(i), d(ii)
(3) a(iii), b(iv), c(i), d(ii) (4) a(iii), b(iv), c(ii), d(i)

[BOTANY]

91. Arrange the following events that occur in sexually reproducing plants in proper sequence and choose the option accordingly.
a. Dehiscence of anther.
b. Use of external agents to achieve pollination.
c. Germination of pollen grains on the stigma of the pistil.
d. Union of two polar nuclei in the embryo sac.
e. Male gametophytes develop inside the microsporangia.
(1) e, a, b, c, d
(2) d, a, c, e, b
(3) d, b, c, e, a
(4) c, e, d, a, b

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92. Read the following statements and choose the option for one(s) that is/are **not** true w.r.t. pollen-pistil interaction.

i. Often, pollen of the wrong type, either from other species or from the same plant also land on the stigma.

ii. In plants that show bud pollination only, self-incompatibility is an important device to enhance genetic variability.

iii. The pollen has the ability to recognise the stigma, whether it is compatible or incompatible.

iv. If pistil accepts the genetically identical pollen, it promotes post-fertilisation events that lead to fertilisation.

v. A special cellular thickenings at the micropylar tip direct the pollen tube growth.

(1) ii, iii and iv

(2) i and v

(3) ii, iv and v

(4) ii only

93. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Each megaspore mother cell in an ovule of most of the flowering plants, divides and form only one megaspore.

Reason (R): Out of the four daughter cells formed after meiosis in megaspore mother cell of the flowering plants, only one survives to form female gametophyte.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) (A) is true but (R) is false

(2) (A) is false but (R) is true

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

94. Which feature of seed is/are crucial for storage of seeds which can be used as food throughout the year and also to raise crop in the next season?

(1) Dehydration and dormancy of mature seeds

(2) Having two generations in a single structure

(3) Tendency to store rich amount of minerals and plant growth regulators

(4) Containing large number of embryos surrounded by seed coat

95. Match the following columns and choose the **correct** option.

	Column I		Column II
A	Dioecy	(i)	Inhibition of pollen germination on flower of same plant
B	Protandry	(ii)	Male and female flowers are present on different plants
C	Hypohydrophily	(iii)	Pollination in submerged female flowers
D	Self-incompatibility	(iv)	Pollen is released before the stigma becomes receptive

(1) A-(iii); B-(ii); C-(i); D-(iv)

(2) A-(iv); B-(i); C-(iii); D-(ii)

(3) A-(ii); B-(i); C-(iv); D-(iii)

(4) A-(ii); B-(iv); C-(iii); D-(i)

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96. A typical angiosperm ovule has one or two protective envelopes called A that encircle the B except at the tip towards where C is organised.

Choose the option to fill in the blanks A, B and C **correctly**.

	A	B	C
(1)	Endothecium	Ovule	Embryo sac
(2)	Sporangial wall	Endosperm	Embryo sac
(3)	Integuments	Nucellus	Egg apparatus
(4)	Integuments	Ovary	Micropyle

97. Which of the following structures in the flowering plants can be formed even if fertilization has **not** occurred?

a. Embryo b. Endosperm
c. Fruit d. Seed

The **correct** one(s) is/are

(1) Only c and d (2) Only a and d
(3) All a, b, c and d (4) Only c

98. Continued self-pollination results in inbreeding depression because

(1) The offspring are genetically identical to the parent as in case of clone
(2) Self-pollination limits the genetic variability in the generations
(3) This phenomenon does not involve crossing over and recombination of genes
(4) Self-pollen become incapable to germinate after few generations

99. Inside the ovary, the ovarian cavity is also called

(1) Locule (2) Carpel
(3) Megasporangium (4) Thecae

100. Given below are two statements:

Statement I: In angiosperms, the seed is a fertilised ovule and it is the final product of sexual reproduction.

Statement II: As the seed matures, its protoplasmic content is increased and the general metabolic activity of the embryo speeds up.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

101. Many insects may consume pollen or the nectar without bringing about pollination. Such floral visitors are referred to as

(1) Pollen or nectar robbers
(2) Predators
(3) Herbivores
(4) Parasites

102. The structure that represents the male gametophyte in flowering plants

I. Does not have cell wall
II. Has vacuolated cytoplasm
III. Possesses germ pores where only plasma membrane is present
IV. Has irregularly shaped nucleus
V. Stores food reserve

The **correct** one(s) is/are

(1) I, II and III (2) II only
(3) IV and V only (4) II, IV and V

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103. Given below are two statements, one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Unlike dicots, seeds of monocots have scutellum.

Reason (R): Seeds of monocot plants need not to store food reserve as that is essential in dicots.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) (A) is true but (R) is false

(2) (A) is false but (R) is true

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

104. To sustain animal visits, the flowers have to provide rewards to the animals. Which of the following is **not** amongst them?

(1) Nectar

(2) Pollen grains

(3) Colourful petals and fragrance

(4) Safe places to lay eggs

105. Read the below given statements and choose the **correct** answer from the following options.

(a) The viability periods of the pollen grains of a species may vary.

(b) In some members of the families to which *Pisum* and *Solanum* belong, pollen grains maintain viability for months.

(c) The basic architecture varies from pollen to pollen within a species of plant.

(d) Pollen grains of a large number of species can be stored for years in liquid carbon dioxide at –196°C.

(e) All the pollen tubes from the germ pores of a pollen grain carry two germ nuclei, each converts into male gametes.

(1) Only statement (c) is incorrect

(2) Statements (c), (d) and (e) are incorrect

(3) Statements (a), (b) and (c) are correct

(4) All the statements are incorrect

106. A typical dicotyledonous embryo consists of two major parts. Identify the statement that is **not** true for this embryo.

(1) Radical having root hairs and root cap is enclosed in coleorrhiza

(2) One of the major parts is cotyledons

(3) Epicotyl of the embryonal axis terminates with the plumule

(4) The cylindrical portion below the level of cotyledons terminates into root tip

107. The species of plants pollinated with biotic agents

I. Can produce seeds that are not enclosed within the fruit.

II. Can be monocotyledonous and dicotyledonous.

III. May produce white fragrant flowers.

IV. May have small flowers clustered into an inflorescence.

The **correct** ones are

(1) Only I and II (2) All I, II, III and IV

(3) Only III and IV (4) Only II, III and IV

108. Identify the following statements as **True (T)** or **False (F)** and choose the option accordingly.

A. The vegetative cell of pollen grain is small and floats in the cytoplasm of the generative cell.

B. When the pollen grain of flowering plants is mature, it mostly contains three cells.

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C. The generative cell of pollen grain is spindle shaped with dense cytoplasm and a nucleus.

D. Male gametophytes of angiosperm are generally spherical measuring about 2-5 micrometers in diameter.

E. In about 40 per cent of angiosperms, the generative cell divides mitotically to give rise to the two male gametes before pollen grains are shed.

	A	B	C	D	E
(1)	F	T	F	T	F
(2)	F	F	T	F	T
(3)	T	F	T	T	F
(4)	F	F	F	T	T

109. Match the following columns and choose the **correct** option.

	Column-I		Column-II
A.	Dormancy	(i)	The wall of fruit that develops from the wall of the ovary
B.	Pericarp	(ii)	The residual, persistent nucellus
C.	Perisperm	(iii)	The cylindrical portion below the level of cotyledons
D.	Hypocotyl	(iv)	A state of inactivity of embryo

(1) A(ii); B(iv); C(i); D(iii)
(2) A(iv); B(iii); C(i); D(ii)
(3) A(iv); B(i); C(ii); D(iii)
(4) A(ii); B(iii); C(iv); D(i)

110. Angiosperm pollen grains are generally spherical with radii about
(1) 25-50 micrometers
(2) 12-25 micrometers
(3) 2.5-5.0 micrometers
(4) 1.2-2.5 micrometers

111. The plant material that cannot be degraded by any enzyme known so far is synthesized by the
(1) Innermost layer of anther wall
(2) Sporogenous tissue of the ovule
(3) Epidermal cells of testa
(4) Cork cambium

112. The mature seeds of groundnut and maize
(1) Do not have membranous seed coat
(2) Contain tissue made up of triploid cells
(3) Are not enclosed in fruit wall
(4) Have the tissue in which cells are filled with reserve food materials

113. Consider the following statements in regards to majority of flowering plant species.
i. They use abiotic agents for pollination.
ii. Pollen grains are shed at three celled stage.
iii. Pollen grains coming in contact with the stigma is only a chance factor.
iv. Insects, particularly bees are the dominant pollinating agents.
The **correct** one(s) is/are:
(1) i and iv
(2) ii only
(3) ii and iii
(4) iv only

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114. Match the following columns and choose the **correct** option.

	Column I		Column II
A.	Stigma	(i)	May be syncarpous or apocarpous
B.	Gynoecium	(ii)	Landing platform for pollen grains
C.	Style	(iii)	The structure through which pollen tube reach the ovary
D.	Pistils	(iv)	May be monocarpellary or multicarpellary

(1) A-(ii); B-(iii); C-(i); D-(iv)
(2) A-(iv); B-(i); C-(iii); D-(ii)
(3) A-(iii); B-(ii); C-(i); D-(iv)
(4) A-(ii); B-(iv); C-(iii); D-(i)

115. White kernel of coconut is
(a) Filled with reserve food materials.
(b) Cellular endosperm.
(c) Used for the nutrition of the developing embryo.
(d) Completely consumed during the embryo formation.

The **correct** one(s) is/are
(1) (b) only (2) (b) and (d) only
(3) (a), (c) and (d) only (4) (a), (b) and (c)

116. Which of the following phenomenon can be seen in plants such as maize?
I. Free nuclear division
II. Geitonogamy
III. Asymmetric cell division
IV. Xenogamy
V. Active movement of male gamete
VI. Degeneration of synergids

Choose the **correct** answer from the options given below.
(1) Only I, II and VI (2) Only III, IV and V
(3) I, II, III, IV and VI (4) Only II, IV and V

117. The process of microsporogenesis in angiosperms does **not** involve
(1) Cytokinesis
(2) Mitosis
(3) Replication
(4) Pairing of homologous chromosomes

118. Free nuclear division in angiosperms occurs
(1) In both haploid and triploid cells of two different generations
(2) Only in diploid cells of reproductive organ
(3) In both diploid and triploid cells of the same generation
(4) Only in triploid cells of post fertilization structure

119. Flowers produce assured seed-set even in the absence of pollinators. This statement can be **true** for which of the following plants?
(1) *Vallisneria* (2) *Carica papaya*
(3) *Zea mays* (4) *Commelina*

120. Read the following statements of Assertion and Reason and in the light of given statements choose the **correct** option.

Assertion (A): Each cell of the sporogenous tissue of an anther is capable of giving rise to a microspore tetrad.

Reason (R): Each cell of the sporogenous tissue of an anther is a potential pollen or microspore mother cell.

(1) (A) is true but (R) is false
(2) (A) is false but (R) is true
(3) Both (A) and (R) are true and (R) is the correct explanation of (A)
(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

Space for Rough Work

121. Read the following statements and choose the **correct** option.

Statement-I: Events of endosperm and embryo development, maturation of ovules into seeds and ovary into fruit, are collectively termed post-fertilisation events.

Statement-II: The cells of endosperm are filled with reserve food materials and are used for the nutrition of the developing embryo sac.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

122. Which of the following statements are **true**?

A. In angiosperms, male gametes are motile whereas female gametes are non-motile.

B. The male and female gametes are brought together for fertilization only by the help of biotic or abiotic agents.

C. Pollination occurs only in those plant groups that form seeds.

D. There is no chance of cross-pollen landing on the stigma in plants such as *Oxalis*.

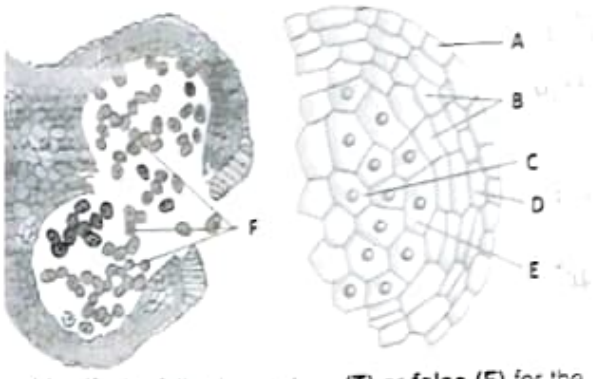
E. In a normal flower which opens and exposes the anthers and the stigma, complete autogamy is rather rare.

Choose the **correct** answer from the options given below.

(1) C, D and E (2) A, B and D

(3) C and E only (4) B, C and E

123. Enlarged view of one microsporangium with wall layers and a part of mature dehisced anther are shown below.



Identify the following as **true (T)** or **false (F)** for the labelled structures and choose the option accordingly.

I. Ploidy of C and F is same and they form the three-celled male gametophyte.

II. A, B and D help in dehiscence of anther.

III. In the mature cells C and E, the number of nuclei per cell in most of the flowering plants remains **the same**.

IV. B is multilayered and D is hygroscopic in *Vallisneria*.

	I	II	III	IV
(1)	T	F	F	T
(2)	F	T	F	F
(3)	F	T	F	T
(4)	T	T	T	F

124. In which of the following cells of a flowering plant, the number of chromosomes is **not** equivalent to that of suspensor cell?

(a) Cell of aleurone layer

(b) Cell of endothecium

(c) Cell of mature tapetum

(d) Cell of cotyledon

Space for Rough Work

(e) Primary endosperm cell

(f) Cell of the nucellus

Choose the **correct** answer from the options given below.

(1) (b), (c) and (d) (2) (a), (c) and (e)

(3) (a) and (e) only (4) (b) and (d) only

125 Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): A typical angiosperm embryo sac, at maturity, though 8-nucleate is 7-celled.

Reason (R): The nuclei of the largest cell of the embryo sac fuse to form a secondary nucleus.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) (A) is true but (R) is false

(2) (A) is false but (R) is true

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

126. Match the following columns and choose the **correct** option.

	Column I		Column II
A.	Anther lobes of a typical angiosperm	(i)	Chalaza
B.	The basal part of ovule	(ii)	Dithecous
C.	Group of cells at the micropylar end of embryo sac	(iii)	Apomixis
D.	Development of seed without syngamy	(iv)	Egg apparatus

(1) A-(iv); B-(i); C-(ii); D-(iii)

(2) A-(ii); B-(iv); C-(i); D-(iii)

(3) A-(ii); B-(i); C-(iv); D-(iii)

(4) A-(iv); B-(iii); C-(i); D-(ii)

127. Regarding double fertilization, which of the following options is **not** true?

(1) Syngamy involves fusion of two gametes, *i.e.*, oogamous type

(2) Disintegration of various haploid cells such as antipodals and vegetative cells

(3) Union of male and female gametes results in the formation of the first cell of next generation

(4) The structure that is formed by the fusion of two polar nuclei with the male gamete gives rise to endosperm

128. In synergids, filiform apparatus

(1) Is special cellular thickenings towards the antipodal cells

(2) Plays an important role in guiding the pollen tubes into the synergid

(3) Is formed by the thinning of cell wall and invagination of plasma membrane into the cell

(4) Provides the condition for division in generative cell to form gametes

129. An ovule inside the ovary in nature is developed into seed having many embryos. These embryos could not be developed from

(1) The cells of inner integument

(2) Cells of nucellus

(3) Cells of pericarp

(4) Egg cell

Space for Rough Work

130. In crossing different species and often genera, to combine desirable characters to produce commercially 'superior' varieties, a breeder generally

(1) Emasculate the male flowers

(2) Bag the emasculated flowers

(3) Cover the stigma of female flower with wax

(4) Remove all the non-essential whorls of hermaphrodite flowers

131. Ploidy of egg apparatus is

(1) 3n (2) n

(3) n + n + n (4) 2n

132. Floral primordia in angiosperms are developed due to

i. Several hormonal activities

ii. Arrival of breeding season of the pollinators

iii. Initiation of structural changes

iv. Scarcity of nutrients in the soil

The **correct** one(s) is/are

(1) Only i and iii

(2) i, ii and iii

(3) Only iii and iv

(4) Only i

133. Some features of a flowering plant **species** could be helpful in artificial hybridization. **Which** of the following is **not** amongst them?

(1) Presence of male and female flowers respectively on distal and proximal parts of a branch of stem

(2) Occurrence of male and female sex organs on two different individuals of the species

(3) Location of anther far apart from the stigma in a flower

(4) Stigma showing compatibility for self-pollen

134. Some events associated with the formation of gametophytic phase in a pistillate flower are given below. Identify them and choose the option accordingly.

i. Megaspore divides mitotically and the daughter ones move to the opposite poles, forming the 2-celled embryo sac.

ii. After the 8-nucleate stage, cell walls are laid down, leading to the organisation of the typical female gametophyte.

iii. Only six of the eight nuclei are surrounded by cell walls and organised into cells.

iv. The two nuclei, called polar nuclei are situated below the egg apparatus in the large central cell.

The **correct** ones are

(1) i, ii and iii

(2) ii, iii and iv

(3) Only i and iii

(4) Only ii and iv

135. The two most important units of sexual reproduction in flowering plants develop in

(1) The same part of a flower situated inside a seed forming structure

(2) Two different parts of a flower situated on the thalamus

(3) The distal prominent end of the pistil

(4) The basal parts of calyx and corolla where nectar is formed

Space for Rough Work

[ZOOLOGY]

136. Consider the statements given below w.r.t. humans.

Statement (A) : The mammary glands of the female undergo differentiation during pregnancy and starts producing milk towards the end of pregnancy by the process called lactation.

Statement (B): After attachment of the trophoblast layer, the uterine cells divide very slowly and cover the morula.

Select the correct option.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is correct

(4) Only statement (B) is correct

137. Monozygotic twins are/have

(1) Different sex always

(2) Formed from the same egg

(3) Share only 50% of the genome

(4) Always connected by two separate placenta to the mother.

138. The functions of male sex accessory ducts and glands are maintained by

(1) Androgen which is secreted by the Leydig cells

(2) Testosterone, a protein hormone, which also acts on the Sertoli cells

(3) FSH which does not act on Sertoli cells

(4) LH which is exclusively responsible for the formation of mammary glands

139. Select the **correct** statement w.r.t. humans.

(1) The ovaries are located one on each side of the upper abdomen.

(2) Spermatogenesis starts at the age of puberty due to a significant increase in the secretion of GnRH.

(3) If the human male ejaculates about 200 to 300 million sperms during a coitus, it indicates azoospermia.

(4) At puberty, only 60,000-80,000 primary follicles are left in both the ovaries.

140. Read the following facts carefully:

A. Anatomically surrounds the first part of the urethra and is present at the base of the bladder.

B. It is an unpaired structure whose alkaline secretion helps to counter the vaginal acidity.

From the options given below, choose the structure for which the above given facts hold true.

(1) Seminal vesicle (2) Prostate gland

(3) Cowper's gland (4) Vas deferens

141. A tertiary follicle is characterized by the presence of

(a) Primary oocyte (b) Granulosa cells

(c) Antrum (d) Theca interna

Choose the **correct** option.

(1) (a) and (b) only (2) (b), (c) and (d) only

(3) (a), (b), (c) and (d) (4) (c) and (d) only

142. Which of the following is not a component of the seminal plasma in humans?

(1) Enzymes (2) Sugar

(3) Sperms (4) Calcium

143. In adult humans, which of the following comparisons between the dimensions of the gonads is correct?

(1) The ovary is generally longer than the testis

(2) The testis is comparatively longer than the ovary

(3) The testis and ovary are equal in length

(4) The ovary is three times wider than the testis

Space for Rough Work

144. If in a woman, ovulation is expected to occur on the 24th day of her menstrual cycle and the luteal phase is of normal duration then, for how many days did the menstrual and proliferative phases last together?
(1) 10 days (2) 14 days
(3) 24 days (4) 28 days

145. The **correct** option w.r.t. the umbilical cord in humans is that it/its
(1) Carries only deoxygenated blood
(2) Development begins in the 16th week of the embryology period
(3) Is the vital connection between the foetus and the placenta
(4) Helps in the removal of only maternal excretory wastes

146. All of the following statements regarding the zona pellucida are correct, **except**
(1) It is a non-cellular glycoprotein layer surrounding the secondary oocyte
(2) It undergoes modification after fertilisation to block the entry of additional sperms
(3) It prevents ectopic pregnancy
(4) It is formed by the primary follicle

147. Which of the following ovarian structures is found exclusively during the luteal phase of a normal menstrual cycle?
(1) Primary follicle (2) Graafian follicle
(3) Corpus luteum (4) Secondary oocyte

148. In a lactating mother, which structure of the mammary gland is primarily responsible for milk synthesis, temporary storage, and is directly surrounded by contractile cells that help in milk ejection?
(1) Ampulla (2) Alveolus
(3) Mammary duct (4) Lactiferous duct

149. Select the option that correctly represents the level of hormones during the onset of parturition in humans.
(1) Estrogen ↑, Progesterone ↑, Prostaglandin ↓
(2) Oxytocin ↑, Estrogen ↓, Relaxin ↑
(3) Cortisol ↓, Estrogen ↑, Progesterone ↓
(4) Oxytocin ↑, Cortisol ↑, Estrogen ↑

150. Choose the option that correctly represents the developments seen in a developing embryo by the end of 24 weeks of pregnancy.

	Limbs with distinct digits visible on ultrasound	Well-differentiated external genitalia	Fine hair covers the body and separated eyelids
(1)	×	√	×
(2)	√	×	×
(3)	√	√	√
(4)	×	√	√

151. The hormones that affects the menstrual cycle in humans is/are
(a) Prolactin
(b) Thyroxine
(c) Melatonin
Select the **correct** option.
(1) (a) only
(2) (b) and (c) only
(3) (c) only
(4) (a), (b) and (c)

Space for Rough Work

152. In a developing mammalian embryo, the layer that ultimately gives rise to structures such as muscles, eyes and heart is

(1) Trophoblast (2) Zona pellucida

(3) Inner cell mass (4) Blastocoel

153. "A 37-years old woman is evaluated for fertility. One of her oocytes is about to complete meiosis-I, just before ovulation".

What can be the approximate age of this oocyte at the time it resumes its meiosis?

(1) Less than 37 years, as its division remained arrested since birth

(2) Only a few months old, as new oocytes are continuously produced after puberty

(3) Approximately 38 years, as it was formed during her foetal life, before birth

(4) Around 25 years, as folliculogenesis occurs only after adolescence

154. Secretions of which of the following are essential for maturation and motility of sperms?

(1) Epididymis, which stores sperms temporarily

(2) Vas deferens, which is directly attached to vasa efferentia

(3) Rete testis, which is directly connected to the vas deferens

(4) Tubuli recti, which is a part of the extra-testicular genital duct system

155. The endometrium is most thickened and richly supplied by arteries (in the span of a 28 days menstrual cycle) when

(1) Estrogen is at its first peak level and the progesterone level is minimum in blood

(2) The corpus luteum has regressed completely

(3) The levels of gonadotrophins are high

(4) Progesterone level is dominant in the secretory phase

156. The correct difference between morula and blastocyst of humans is that the

(1) Latter contains only 4 – 5 totipotent blastomeres

(2) Former is a hollow ball of 64 cells

(3) Latter is formed within the uterus

(4) Former has more DNA content than the latter

157. A modulator molecule that enhances the estrogen's positive feedback to hypothalamus is given to a woman during her follicular phase, then this modulator will

(1) Decrease FSH secretion due to high level of GnRH

(2) Cause menstrual flow immediately

(3) Induce ovulation due to the abnormal surge of progesterone and LH

(4) Increase the secretion of GnRH, FSH and LH, leading to excessive follicular stimulation

158. **Assertion (A):** The anatomical location of testes in human males is an adaptation linked to gamete production.

Reason (R): Efficient spermatogenesis requires the temperature which is slightly higher than the normal core body temperature.

In the light of above statements, choose the **correct** option.

(1) Both (A) and (R) are true, (R) explains (A) correctly

(2) Both (A) and (R) are true, (R) does not explain (A) correctly

(3) (A) is true (R) is false

(4) (A) is false (R) is true

Space for Rough Work

159. A fully mature follicle, just prior to ovulation, is precisely
(1) Located deep within the ovarian cortex, which is the inner part of the ovary
(2) Embedded in the ovarian medulla, near the vascular elements
(3) Present at the cortico-medullary junction of the ovary
(4) Present immediately beneath the surface of the ovarian epithelium

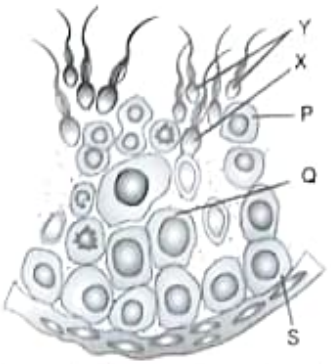
160. Read the following statements w.r.t. the fallopian tube of humans:
(a) It is a part of the female sex accessory duct.
(b) It extends from the periphery of each ovary to the uterus.
(c) It is lined by the type of epithelium that contains spherical and centrally located nucleus.
(d) It's middle part is called ampulla which opens directly into the abdominal cavity.
Select the **correct** option.
(1) Statements (c) and (d) are correct
(2) Statement (b) is incorrect, but (a), (c) and (d) are correct statements
(3) Statements (a) and (b) are correct, but (c) and (d) are incorrect statements
(4) Statement (b) and (d) are incorrect, but (a) and (c) are correct statements

161. Choose the **correct** match w.r.t. the male reproductive system.

(1)	Sertoli cells	–	Provide nutrition to the germ cells
(2)	Interstitial spaces of testis	–	Contain only immunologically competent cells
(3)	Glans penis	–	Loose fold of skin covering the enlarged end of penis
(4)	Rete testis	–	Part of extra-testicular genital duct system

162. All of the following are **correct** w.r.t. the testes of humans, **except**
(1) Paired oval structures
(2) Are covered by a dense fibrous connective tissue layer
(3) Have about 250 testicular lobules in total
(4) Are the primary sex organs

163.



Analyse the image given above that represents the sectional view of a seminiferous tubule. Identify the labelled cells and choose the **incorrect** option w.r.t. them.
(1) 'S' is a stem cell that lies in contact with the basement membrane of the seminiferous tubules.
(2) Transformation of 'X' into 'Y' is termed as spermiation.
(3) 'P' is a haploid cell.
(4) 'Q' undergoes the first meiotic division, leading to the formation of two 'P' cells.

164. In human females, menopause is a change which
(1) Occurs generally around the age of 50 years
(2) Is characterized by highly elevated levels of estrogen
(3) Is marked by increased calcification of bones
(4) Is characterized by extremely low levels of gonadotropins

Space for Rough Work

165. Comprehend the statements given below w.r.t. the menstrual cycle of a human female:
(a) The pre-ovulatory phase of the cycle is more variable in length than the other phases.
(b) The high levels of estrogen during the pre-ovulatory phase exert only negative feedback effect on LH.
(c) The follicular phase (in the ovary) is preceded by the proliferative phase (in the uterus).
(d) Both FSH and LH attain a peak level in the middle of the cycle *i.e.*, on the 14th day of the 28 days cycle.
How many of the above statements is/are correct?
(1) One (2) Three
(3) Two (4) Four

166. The parts of the female reproductive system along with the mammary glands are integrated both structurally and functionally to support how many processes given in the box below?

Ovulation, Fertilisation, Pregnancy, Birth, Child care

Choose the **correct** option.
(1) Four (2) Three
(3) Two (4) Five

167. Which chromosomal combination is essential for the development of a normal female child?
(1) 44 + XO (2) 46 + XX
(3) 44 + XY (4) 44 + XX

168. Choose the **correct** statement.
(1) The 1st and 2nd polar bodies are different on the basis of number of chromosomes.
(2) In humans, the number of mammary tubules are more than the number of mammary lobes.
(3) In human females, the infundibulum leads to a narrower part of the oviduct, called ampulla.
(4) Monkeys and apes do not exhibit menstrual cycle.

169. Match **column I** and **column II** w.r.t. humans.

	Column I		Column II
(a)	Nebenkern	(i)	Contains hydrolytic enzymes
(b)	Perimetrium	(ii)	Located in the middle piece of the sperms
(c)	Acrosome	(iii)	Embryo formation
(d)	Trophoectoderm	(iv)	Outermost serosal layer of the uterus

Choose the correct option.

	(a)	(b)	(c)	(d)
(1)	(i)	(ii)	(iv)	(iii)
(2)	(ii)	(iv)	(i)	(iii)
(3)	(ii)	(i)	(iv)	(iii)
(4)	(ii)	(i)	(iii)	(iv)

170. Select the **incorrect** match w.r.t. humans.

(1)	Cervical canal	–	Cavity of the cervix
(2)	Perivitelline space	–	Space between the zona pellucida and cell membrane of an oocyte
(3)	Urethral meatus	–	Common exit passage for gamete and urine in both the sexes
(4)	Amnion	–	Extra-embryonic membrane

Space for Rough Work

171. Comprehend the table given below w.r.t the hormonal profile of an early pregnancy in an adult female.

Hormones	Plasma levels of hormones
hCG	↑
FSH	↓
LH	↓

Based on the given data, we can correctly infer that

(1) ↑ hCG is seen due to the degenerated corpus luteum

(2) ↓ LH is observed due to the negative feedback control of GnRH, which is secreted by the anterior pituitary

(3) ↑ hCG maintains the corpus luteum which synthesises of progesterone and estrogen

(4) ↓ FSH level is seen as no new ovarian follicles are formed during pregnancy due to the positive feedback of estrogen on the hypothalamus.

172. Consider the statements given below w.r.t humans:

Statement A: When a female child is born, each primary oocyte is surrounded by a single layer of granulosa cells.

Statement B: If the secondary oocyte is not fertilized about two days before the end of the monthly cycle, the corpus luteum in the ovary involutes and the level of ovarian hormones increases in the blood plasma.

Select the **correct** option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is incorrect

(4) Only statement B is incorrect

173. Select the event that does not exclusively occur in females.

(1) Capacitation (2) Insemination

(3) Gestation (4) Gametogenesis

174. **Assertion (A):** The extrusion of the second polar body is temporarily coupled with fertilization, that ultimately leads to the formation of a haploid zygote.

Reason (R): The secondary oocyte, arrested at anaphase-II, undergoes a reductional division only after the entry of the sperm to complete meiosis-II, leading to syngamy and formation of a single-celled zygote.

In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true, (R) correctly explains (A)

(2) Both (A) and (R) are true, (R) does not explain (A) correctly

(3) (A) is true; (R) is false

(4) Both (A) and (R) are false

175. The human milk consists of all of the following, **except**

(1) Fat, in the form of globules

(2) Lactose, a milk sugar

(3) IgA, a homopolymer

(4) Mineral salts, such as Ca^{+2}, Na^{+}, *etc.*

176. The component/structure which is absent in ovum, but present in sperm is

(1) Mitochondria

(2) Cytoplasm

(3) Plasma membrane

(4) Flagellum

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177. Select the **incorrect** option w.r.t. uterus.
(1) Also known as womb
(2) Contributes in placenta formation
(3) Takes part in expulsion of the baby during parturition
(4) Supported primarily by tendons attached to the pelvic wall

178. Consider the facts given below w.r.t. a part of the female reproductive system:
(a) Located at the upper junction of labia minora
(b) Tiny finger-like structure which is composed of erectile tissue
(c) A part of the external genitalia
All of the above given facts are true for
(1) Clitoris
(2) Labia majora
(3) Mons pubis
(4) Hymen

179. Choose the **incorrect** option w.r.t. hormones produced by the ovaries in females.
(1) Inhibin, inhibits the secretion of FSH
(2) Estrogen, responsible for the development of secondary sexual characteristics
(3) Progesterone, maintains the uterine lining for implantation
(4) Relaxin, steroidal hormone secreted by the granulosa cells to relax the pelvic joints during childbirth

180. In human males, the ___ dilates to form ampulla, which receives the duct from the ___ behind the urinary bladder and forms an ejaculatory duct. Select the **correct** option to fill the respective blanks.
(1) Epididymis; Urethra
(2) Seminal vesicle; Ductus deferens
(3) Rete testis; Vasa efferentia
(4) Vas deferens; Seminal vesicle

❑ ❑ ❑

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